Applicant: LUO Xiaoyu
Program: MSc Intelligent Robotics Engineering, The Hong Kong Polytechnic University

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PERSONAL STATEMENT

From hardware coordination to software architecture practices, I firmly believe that the ultimate mission of technology is to build reliable, efficient systems that can solve real-world problems. This belief has been driving me to constantly ask: Can the system autonomously respond in dynamic scenarios with unclear rules? And can it extract information from real, unstructured physical interactions and make intelligent decisions? After witnessing the break through happening in the field of artificial intelligence, I now realize that artificial intelligence provides the core engine for solving problems, while intelligent robots are the bridge that injects its value into the physical world. However, I also perceive that building this bridge requires crossing the critical gap from algorithms to entities. This is precisely the core goal I am eager to achieve in the MSc Intelligent Robotics Engineering program at The Hong Kong Polytechnic University. I aim to master the complete technical stack of robot perception, cognition, and control.

My academic journey began with the interdisciplinary exploration of computer systems and the principles of artificial intelligence. Through the algorithmic cognition acquired from *Data Structures and Algorithms*, I realized that the core of robot intelligence lies in converting perceptual data into a computational pipeline for intelligent decision-making. The course Software System Design and Architecture cultivated my ability to build complex modular systems, which is directly applicable to the design and integration of robot functional nodes. Based on this, I realized that whether it is path planning based on search algorithms or environment understanding based on machine learning, the underlying mechanism relies on precise control of computational complexity and real-time performance. At the same time, the course Operating Systems Principles enabled me to master the core mechanisms of resource scheduling and process management, providing a theoretical foundation for understanding real-time task processing and hardware coordination in robot systems. The Robot Technology course allowed me to apply theory to practice for the first time. During this process, by enabling the mobile robot to achieve autonomous navigation and automatic path planning, I deeply realized that path planning is not only about seeking the optimal solution but also needs to take into account sensor noise, motion dynamics and real-time environmental interaction. A reliable robot system must strike a balance between algorithmic intelligence and physical limitations..

Beyond the classroom, I continued to deepen my knowledge and aspirations through project practice. I focus on developing engineering skills for solving complex system problems. When developing the traffic query system, I quickly learned the Vue framework and mastered the methodology of decomposing complex interfaces into components. This system decomposition and integration thinking is equally crucial in the software architecture of robots. Moreover, when dealing with the real challenges encountered during the server operation internship. The automatic migration script I designed failed due to the dynamic update of backup files. Viewing this, through calling the cloud API to dynamically query the server status, I endeavored to investigate the problem and solve it effectively. This experience enabled me to deeply understand the importance of Real-time state estimation in dynamic environments and adaptive decision-making. I believe this is similar to the principle that robots use sensor information to make real-time plans in uncertain environments. Besides, an incident of mistakenly deleting configuration settings gave me a profound and unforgettable understanding of the security boundaries and system reliability of the production environment. Under the guidance of a senior engineer, I endeavored to utilize methodology of root cause analysis through segmented logs. This experience not only honed my rigor and composure when dealing with complex systems but also enabled me to establish an engineering methodology that is problem-oriented and solves problems through iteration and verification. I think it is crucial for my future work in developing high-risk robotic systems.

These experiences of dealing with complex systems in real engineering environments have enabled me to deeply understand the boundaries of my knowledge system. I can solve specific problems through dynamic perception and adaptive decision-making, and also locate the root cause of faults through systematic troubleshooting. However, the solutions I possess are still scattered and based on specific scenarios. It is this perception that triggers me to seek a profound academic

learning targeted to the improvement of these deficiencies. I am eager to seize the opportunity to deeply study at the MSc Intelligent Robotics Engineering at The Hong Kong Polytechnic University.
My application stems from a very specific and clear demand. I need to utilize the course system of your program to integrate my scattered knowledge, and at the same time, transform my practical experience in dealing with system complexity into the theoretical foundation for designing an inherently reliable robot system. I plan to focus on studying the two courses: *Principles of Robotic Mechanisms* and *Industrial Human-Robot Systems and Automation*.

During the operation and maintenance process, although I could sense the server status through the API, I was unable to execute physical repairs proficiently. This has made me deeply realize that intelligence without mechanical principles as a foundation is incomplete. I intend to investigate the course *Principles of Robotic Mechanisms* to study kinematic modeling, the characteristics of transmission mechanisms and dynamic analysis. I hope to transform my proficient perception and decision-making algorithms into precise, stable and efficient actions of the robot's end effector in uncertain environments. Apart from this, I once mistakenly deleted the configuration due to improper permission settings. This experience has left me with a profound understanding of security boundaries. I believe that this lesson is in perfect alignment with the core of *Industrial Human-Robot Systems and Automation:* to define an absolutely reliable safety boundary for intelligent agents in a shared space. I am eager to systematically study the ISO/TS 15066 safety standards and the active obstacle avoidance strategy based on torque control and visual monitoring. I hope to transform my passive experience in handling system failures into the forward-looking ability to design inherently safe human-machine systems.

Last but not least, I also notice and favor the humanoid robot platform, the high-precision industrial robotic arm cluster, and the drone experimental matrix that this program possesses. These technological advantages will greatly enable me to verify the algorithms in a real physical environment, rather than just through simulations. I aspire to utilize these facilities to study the real-time scheduling issues in multi-robot collaborative operations.

In a nutshell, I aim to extend the distributed system experience I have gained in server operation to the collaborative control of physical robots. After graduation, I aim to join the leading robotics research team in the industry, dedicated to converting cutting-edge algorithms into industrial-level solutions and applying them in scenarios such as precise assembly and intelligent logistics.

实习证明

兹有华南师范大学软件工程专业罗霄羽同学，身份证号为430922200408089613，于2025年7月2日至2025年9月30日在我司进行实习， 其实习岗位是: 技术研发部实习生。 该同学在实习期间主要参与了以下工作:

一、协助开发二进制模糊测试agent的开发，负责基础业务框架的构建以及模糊测试过程大语言模型的应用。

二、参与llm驱动符号化漏洞检测器的开发，辅助进行漏洞信息搜集以及通用大模型解析效率的对比实验。

该同学具备积极的学习态度、良好的团队协作能力，并在专业知识的深化理解、实际操作技能及问题解决能力方面取得显著进步。

特此证明。

单位名称：

时间：2025.10.14

Internship Certificate

This is to certify that Mr. LUO Xiaoyu, ID no. 430922200408089613, majoring in Software Engineering at South China Normal University, interned as Technical R&D Intern at our company from July 2, 2025, to September 30, 2025. During the internship, he mainly engaged in the following tasks :

1. Assisted in developing a binary fuzzing test agent, responsible for constructing the foundational business framework and implementing large language model (LLM) applications in the fuzzing process.

2. Contributed to the development of an LLM-driven symbolic vulnerability detector, supporting vulnerability information collection and comparative experiments on parsing efficiency of general-purpose large models.

Mr. LUO demonstrated a proactive learning attitude and strong teamwork skills. Through this internship, he significantly enhanced his technical expertise, practical problem-solving abilities, and operational proficiency.

Hereby Certify.

Company:

(Stamp)

Date: 2025.10.14

广州网律互联网科技有限公司

实习证明

兹有华南师范大学软件工程专业罗霄羽同学，身份证号为430922200408089613，于2024年10月29日至2024年12月12日在我司进行实习， 其实习岗位是：技术研发部实习生。 该同学在实习期间主要参与了以下工作：

一、协助开发内部坐席管理系统，负责异常处理部分模块的代码编写、测试和优化。

二、参与公司内部软件系统的维护和升级，解决了堡垒机服务器自动化漂移脚本及开机自检脚本编写等技术问题。

纵观他实习期间的表现，他能够积极学习新的计算机技术和知识，与团队成员合作良好，展现出较强的学习能力和团队协作精神。通过实习，该同学对计算机专业知识有了更深入的理解和掌握，具备了一定的实际操作能力和问题解决能力。

特此证明。

公司名字：广州网律互联网科技有限公司

联系电话：020-88801242

公司盖章：

负责人签字：

日期：2024年12月12日

Guangzhou Wanglv Internet Technology Co., Ltd.

Internship Certificate

This is to certify that Mr. LUO Xiaoyu, ID no. 430922200408089613, majoring in Software Engineering at South China Normal University, interned as intern at the Technical R&D Department of our company from October 29, 2024 to December 12, 2024. During the internship, he mainly engaged in the following tasks :

1. Assisted in developing the internal agent management system, and underwent the coding, testing, and optimization of the exception handling module.
2. Participated in maintaining and upgrading internal software systems, resolving technical issues including automated drift scripting for JumpServer and startup self-check script development.

Based on his performance, I found that he could take the initiative to learn new computer technologies and knowledge, and cooperated well with team members, demonstrating strong learning ability and teamwork spirit. Through the internship, Mr. LUO Xiaoyu has gained a deeper understanding and mastery of computer-related knowledge, and acquired certain practical operation skills and problem-solving abilities.

Hereby Certify.

Company: Guangzhou Wanglv Internet Technology Co., Ltd.

Contact Phone: +86 20 3880 1242

Company Seal:

Signature of Authorized Person: chenli

Date: December 12, 2024

实习证明

兹有华南师范大学软件工程专业罗霄羽同学，身份证号为：430922200408089613，于 2024 年 7 月 1 日至 2024 年 8 月 31 日在我司进行实习， 其实习岗位是：IT 开发助理工程师。 该同学在实习期间主要参与了以下工作：

一、协助开发配置报价系统，负责部分模块的代码编写、测试和优化。

二、参与公司配置报价系统的维护和升级，完成了梯度管理、参数配置、汇率管理等业务模块开发。

纵观他实习期间的表现，他能够积极学习新的计算机技术和知识，与团队成员合作良好，展现出较强的学习能力和团队协作精神。通过实习，该同学对计算机专业知识有了更深入的理解和掌握，具备了一定的实际操作能力和问题解决能力。

特此证明。

单位名称：深圳麦克韦尔科技有限公司

时间：2024 年 8 月 31 日

深圳麦克韦尔科技有限公司 人力资源专用章

Internship Certificate

This is to certify that Mr. LUO Xiaoyu, ID no. 430922200408089613, majoring in Software Engineering at South China Normal University, interned as IT Development Assistant Engineer at our company from July 1, 2024 to August 31, 2024. During the internship, he mainly engaged in the following tasks :

1. Assisted in the development of Configuration quotation system, and took control of writing, testing and optimizing some modules of the code.

2. Participated in the maintenance and upgrade of the company's configuration quotation system, and completed the development of business modules such as gradient management, parameter configuration, and exchange rate management.

Based on his performance during the internship, he could take the initiative to learn new computer technologies and knowledge, and cooperated well with team members, demonstrating strong learning ability and teamwork spirit. Through the internship, Mr. LUO Xiaoyu has gained a deeper understanding and mastery of computer-related knowledge, and acquired certain practical operation skills and problem-solving abilities.

Hereby Certify.

Company: Smoore International Holdings Limited

Date: August 31, 2024

深圳麦克韦尔科技有限公司
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